Final Project Report: Sleep Patterns and Quality Analysis

1. Introduction

Over the course of this term, I conducted a self-tracking project focused on my sleep habits. The goal of this project was to explore the relationship between sleep duration, bedtime, and overall sleep quality. Sleep is a critical factor in physical and mental well-being, and I wanted to better understand how my own behaviors impact the quality of my rest.

This project combines personal data collection with data visualization techniques using D3.js. By analyzing patterns in my sleep data, I aimed to uncover trends, test a hypothesis, and reflect on how lifestyle habits influence sleep outcomes.

2. Data Collection

I collected sleep-related data daily over multiple weeks. Each entry represents one night of sleep and includes the following variables:

- **Date** – The specific day the data was recorded
- **Bedtime** – The time I went to bed
- **Wake-up Time** – The time I woke up
- **Hours Slept** – Total hours of sleep (calculated)
- **Time to Fall Asleep (Minutes Awake)** – Time spent in bed before sleeping
- **Sleep Quality (1–5)** – Subjective rating of sleep quality
- **Notes** – Additional observations (e.g., stress, environment, interruptions)

Link to Data Collection Table

https://docs.google.com/document/d/13Ga_wh09-qrpLTFwVb5DEziiHX4KuEX4Bl3OkjabdRE/edit?usp=sharing

Data Adjustments

During the project, I refined how I recorded:

- Time-to-fall-asleep (more precise tracking)

- Sleep quality (more consistent scale interpretation)

These adjustments improved consistency and reliability of the data.

3. Research Question and Hypothesis

Research Question

How do sleep duration and bedtime affect sleep quality?

Hypothesis

As my daily hours of sleep and bedtime consistency increases, my sleep quality increases.

4. Visualization Design

To explore this question, I created a **bubble (scatter) plot** using D3.js.

Why this visualization?

A scatterplot is ideal because:

- It shows the relationship between two quantitative variables (hours slept and sleep quality)
- It allows patterns and correlations to be visually identified
- It can be extended with additional variables (bubble size and color)

Encoding Choices

Variable	Visual Encoding
Hours Slept	X-axis
Sleep Quality	Y-axis
Minutes Awake	Circle size (radius)
Bedtime Category	Circle color

Bedtime Categories

- Before 11 PM → Blue
- 11 PM – 1 AM → Yellow
- After 1 AM → Red

This added a categorical dimension that helps reveal patterns beyond just hours slept.

5. How the Visualization Works

The visualization plots each night of sleep as a circle:

- The **position** shows how long I slept and how good the sleep was.
- The **size** represents how long I was awake before I fell asleep.
- The **color** represents bedtime category.
- **Tooltips** provide detailed information when hovering.

To improve readability:

- I used **jitter** to reduce overlap between points.
- I **sorted data by size**, so smaller circles appear on top.
- I fixed axis ticks to show only meaningful values.

6. Findings and Observations

From analyzing the visualization, several patterns emerged:

1. Positive relationship between sleep duration and quality

There is a general trend where higher hours of sleep correspond to higher sleep quality. Nights with 7–9 hours of sleep tend to have quality ratings of 4 or 5.

2. Late bedtimes reduce sleep quality

Points in the “after 1 AM” category (red) tend to cluster in lower quality values. This suggests that staying up late negatively affects sleep.

3. Time to fall asleep matters

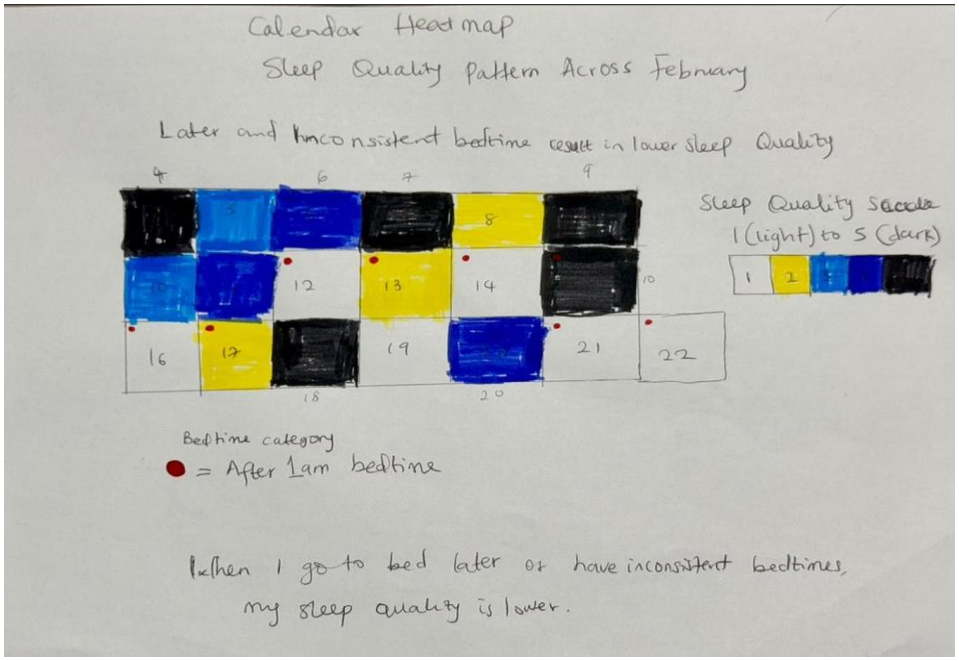
Larger circles (more minutes awake) are often associated with lower sleep quality. This indicates that difficulty falling asleep reduces overall restfulness.

4. Inconsistency impacts results

Some nights with enough sleep still have lower quality scores. Notes suggest this may be due to stress, distractions, or irregular routines.

7. Alternative Visualization Considered

I considered using a **Heat map** to show sleep trends over time.



Benefits:

- Shows patterns across days
- Highlights consistency or irregularity

Drawbacks:

- Harder to compare multiple variables simultaneously
- Does not effectively show relationships between sleep duration and quality

Final Decision:

I chose the **scatterplot** because it better answers the research question by directly comparing variables.

8. Design Rationale

The final visualization was chosen because it:

- Clearly shows relationships between variables
- Supports multiple data dimensions (size, color, position)
- Allows for easy interpretation of patterns

I also referenced general data visualization principles:

- Use scatterplots for relationships
- Use color for categorical variables
- Use size for magnitude

9. Challenges and Iterations

During development, I encountered several issues:

1. Overlapping circles

- Problem: Small circles were hidden behind larger ones
- Solution: Sorted data so larger circles draw first

2. Axis readability

- Problem: Too many ticks cluttered the axes
- Solution: Limited ticks to specific values

3. Legend alignment

- Problem: Labels and circles were misaligned
- Solution: Adjusted spacing and positioning

4. Data categorization errors

- Problem: Some bedtimes were misclassified
- Solution: Improved conditional logic

These iterations improved both functionality and visual clarity.

10. Self-Analysis and Reflection

What I expected

I expected that more sleep would lead to better quality and that earlier bedtimes would help.

What I learned

The data supported my hypothesis, but also revealed:

- Consistency matters as much as duration
- Falling asleep quickly is important
- External factors (stress, habits) influence results

Surprises

Some nights with enough sleep still had poor quality, showing that sleep is influenced by more than just time.

Impact on me

This project made me more aware of my habits. I now:

- Pay attention to bedtime consistency
- Avoid distractions before sleep
- Reflect more on daily routines

11. Future Improvements

If I continued this project, I would collect:

- Screen time before bed
- Stress levels
- Physical activity
- Diet or caffeine intake

This would allow deeper analysis of causes and effects.

12. Reflection on Data Tracking

Before this project, I did not consistently track personal data. This experience showed me:

- The value of data in understanding behavior
- How visualization reveals patterns not obvious in raw data

I am now more interested in tracking habits and using data to improve daily life.

13. Works Cited

- Munzner, Tamara. *Visualization Analysis and Design*. CRC Press.
- "From Data to Viz." https://www.data-to-viz.com
- Taylor-Laird, Jay. Course lectures and materials.
- D3.js Documentation. https://d3js.org

14. Additional Resources

- Tutorials on D3 scales and axes
- Examples of scatterplots and bubble charts
- Online references for tooltip interaction